MIL-S-52432A
26 July 1985

SUPERSEDING
MIL-S-52432
19 July 1965

MILITARY SPECIFICATION

SLINGS, MULTIPLE LEG (CHAIN)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers multiple-leg chain slings with hooks for attachment to loads.

1.2 Classification. Slings shall be of the following classes, as specified (see 6.2):

Class 1 - Pipe sling
Class 2 - Crane sling
Class 3 - Special purpose sling
Class 4 - Drum sling
Class 5 - Fire control equipment sling
Class 6 - Mine sling
Class 7 - Logging sling

1.2.1 Specification definitive part number. The specification part number is definitive and formatted to identify each item covered by this specification. The part number is formatted by selecting from the requirement options available in this specification as follows (see 3.7):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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Definitive Specification Part Number M 52432 X XXX

M Prefix - an item defined by inch-pound (US) units
Specification number
Class of sling required (see table I)
Leg length required in inches (see table II)

Example: M52432F109 - a Class 6 sling with 109-inch legs.

1.2.1.1 Class of sling designator. A one-position alpha field used to designate the class of sling required (see table I).

TABLE I.

Dash no.	Class	Description of sling
A	1	Pipe sling
B	2	Crane sling
C	3	Special purpose sling
D	4	Drum sling
E	5	Fire control equipment sling
F	6	Mine sling
G	7	Logging sling

1.2.1.2 Length of leg designator. A three-position numeric field used to designate the length of the sling legs in inches. Classes 1 through 5 have single standard lengths, while classes 6 and 7 have three lengths each (see table II).

TABLE II

Dash no.	Leg length (Inch)	Class
060	60	1
360	360	2
125	125	3
021	22	4
180	180	5
065	65	6
071	71	6
109	109	6
039	39	7
046	46	7
054	54	7

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-766	- Steel Plates, Sheets, and Strip-Corrosion Resisting.
QQ-W-423	- Wire, Steel, Corrosion-Resisting.
RR-C-271	- Chains and Attachments, Welded and Weldless.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The multiple leg chain slings, hereinafter referred to as "sling", shall be as shown in figures 1 through 4 and as specified herein.

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Material deterioration and control. The sling shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the sling may be exposed.

3.2.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.2.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

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3.2.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components and sub-components incorporated in the sling may be newly fabricated from recovered materials to the maximum extent practicable, provided the sling produced meets all other requirements of this specification. Used, rebuilt or remanufactured components and sub-components shall not be incorporated in the sling.

3.2.3 Detailed material requirements.

3.2.3.2 Material finish. All components and sub-components of the sling that are not corrosion resistant shall be hot-dipped galvanized in accordance with ASTM A153.

3.2.3.3 Components and sub-components.

3.2.3.3.1 Chain. Chain shall be in accordance with RR-C-271, type I, grade C, class 1, zinc coated.

3.2.3.3.2 Lifting rings. Lifting rings shall be in accordance with RR-C-271, type VI.

3.2.3.3.3 Hoist hooks. Hoist hooks shall be in accordance with RR-C-271, type V, class 3.

3.2.3.3.4 Slip hooks. Slip hooks shall be in accordance with RR-C-271, type V, class 1.

3.2.3.3.5 Logging hooks. The material of the logging hooks shall be grade C steel in accordance with RR-C-271.

3.2.3.3.6 Drum hooks. The material of the drum hooks shall be grade C steel in accordance with RR-C-271.

3.2.3.3.7 Anchor shackles. Anchor shackles shall be in accordance with RR-C-271, type IV, class 1, except that the dimensions shall be as specified in 3.5.

3.2.3.3.8 Chain shackles. Chain shackles shall be in accordance with RR-C-271, type IV, class 5.

3.2.3.3.9 Connecting links. Connecting links shall be in accordance with RR-C-271, type II.

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5	1,700	3,400
6	932	2,300
7	7,950	15,900

1/ pounds

3.5 Detailed requirements.

3.5.1 Class 1 and 2 slings. The class 1 and 2 slings shall have 2 legs with a lifting ring commonly attached to one end of each leg and hoist hook attached to the other end of each leg. The legs shall be attached to the ring and the hooks by a connecting link.

3.5.2 Class 3 sling. The class 3 sling shall be as specified in 3.5.1, except that the hooks shall be slip hooks.

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3.5.3 Class 4 sling. The class 4 sling shall be as specified in 3.5.1, except that the lifting ring shall be replaced by a pear shaped link as shown in figure 5 and the hooks shall be drum hooks as shown in figure 6.

3.5.4 Class 5 sling. The class 5 sling shall be as specified in 3.5.1, except that the lifting ring shall be replaced by a pear shaped link as shown in figure 5.

3.5.5 Class 6 sling. The class 6 sling shall be as specified in 3.5.1, except that it shall have 3 legs.

3.5.6 Class 7 sling. The class 7 sling shall have 3 legs commonly attached to an anchor shackle on one end of each leg and terminated with a logging line hook on the other end of the leg. Attachment of the legs to the anchor shackle shall be with pear shaped links as shown in figure 5 and attachment of the logging hooks shall be with chain shackles.

3.6 Dimensions. The dimensions of the sling legs, components and sub-components of the sling shall be as shown in table IV.

3.6.1 Leg length tolerance. The length of the sling legs shall be not more than 4 percent longer or shorter than the leg lengths specified in table IV.

3.7 Identification marking. The sling shall be identified in accordance with MIL-STD-130. A corrosion resistant steel tag in accordance with QQ-S-766, class 301, 302, or 304, annealed condition, number 1 finish, not less than 0.016-inch thick shall be securely attached to the sling with a corrosion-resisting wire in accordance with QQ-W-423, form 1, composition 302 or 304, condition A. The tag shall have the following information embossed, etched, or stamped thereon:

U.S. (Specification part number)
Class _____: (Name of class) Sling
Rated capacity: _____pounds (Safe working load)
Contract No. _____

3.8 Workmanship. The sling shall be manufactured and assembled so that the sling and its components and sub-components are free from cracks or fractures (see 6.3), corrosion, fins, burrs, seams or sharp edges. Screw threads shall be true to form, clean cut, and free from nicks or distortions.

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TABLE IV. Dimensions of sling, components, and sub-components.

Sling class	Chain size (Inch)	Leg length (Inches)	Lifting ring size (Inches)	Hoist hook chain size (Inch)	Slip hook chain size (Inch)	Logging line hook size	Drum hook size	Anchor shackle size	Chain shackle size (Inch)	Connecting link size (Inch)	Pear shaped link (Fig. 5)
1	3/8	60	5/8 x 3	3/8						1/2	
2	1/2	360	7/8 x 3-1/2	1/2						3/4	
3	5/8	125	1-1/8 x 5		5/8					7/8	
4	3/8	22					Fig. 6			1/2	Class 4
5	3/8	180 109		5/16						1/2	Class 4
6	3/8	71 65	5/8 x 3	3/8						1/2	
7	7/8	54 46 39				Fig. 4		Fig. 4	1		Class 7

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).

b. Inspection of packing (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examinations shall be in accordance with MIL-STD-105.

4.3.2 Examinations.

4.3.2.1 Individual. Each sling shall be visually examined for the critical defect specified in 4.4.1. Presence of a critical defect shall be cause for rejection.

4.3.2.2 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in 4.4.1. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

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4.3.3 Tests. Each sling shall be tested as specified in 4.4.2. The contractor shall retain a certificate of the proof test and shall make it available for examination by the contracting officer or designated representative upon request.

4.4 Inspection procedure.

4.4.1 Examination. The slings shall be examined for the following defects:

Critical

1. Fractures in any chain link, lifting ring, hook, shackle, or link.

Major

101. Materials not as specified.
102. Materials not resistant to corrosion and deterioration or not treated to be made resistant to corrosion and deterioration.
103. Dissimilar metal not in accordance with MIL-STD-889.
104. Material and material finish or treatment information not available.
105. Used, rebuilt or remanufactured components or sub-components incorporated in the sling.
106. Components or sub-components, that are not corrosion resistant, not galvanized.
107. Chain not as specified.
108. Lifting rings not as specified.
109. Hoist hooks not as specified.
110. Slip hooks not as specified.
111. Logging hooks not as specified.
112. Drum hooks not as specified.
113. Anchor shackles not as specified.
114. Chain shackles not as specified.
115. Connecting links not as specified.
116. Pear shaped links not as specified.
117. Design of sling not as specified.
118. Performance and safe working load not as specified.
119. Class 1 and 2 slings not as specified.
120. Class 3 slings not as specified.
121. Class 4 slings not as specified.
122. Class 5 slings not as specified.
123. Class 6 slings not as specified.
124. Class 7 slings not as specified.
125. Dimensions and tolerance of sling legs not as specified, or dimensions of components and sub-components not as specified.
126. Workmanship not as specified.

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Minor

201. Identification marking not as specified.

4.4.2 Test. The test may be performed on a hydraulic testing machine having a loading capability equal to the proof load of the sling as specified in 4.4.2.1, or by a lifting device with a safe working load equal to the proof load of the sling and a deadweight load equal to the proof load of the sling as specified in 4.4.2.2.

4.4.2.1 Hydraulic proof test. The top attachment of the sling (lifting ring for class 1, 2, 3, and 6, pear shaped link for class 4 and 5, and anchor shackle for class 7 slings) shall be attached to the dead end of the hydraulic testing machine. The hooks on the hook ends of the legs shall be attached to the moving end of the machine by links or connectors that will distribute the stress to all legs equally. The proof load specified in table I shall be applied.

4.4.2.2 Lifting proof test. The top attachment (see 4.4.2.1) shall be attached to the lifting device. The hooks on the hook ends of the legs shall be attached to the proof load. The lifting device shall be activated until the proof load is lifted clear of the supporting surface on which it was resting.

4.4.2.3 Test failure. After the sling has been stressed by the proof load, release the load, remove the sling and examine it. Nonconformance to 3.4 shall constitute failure of the test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

No.	Defect	A	B	Comm
127.	Slings not preserved as specified.	5.1.1		5.1.2
128.	Shipping container not as specified.	5.2.1	5.2.2	5.2.3
129.	Unlike slings packed in same box.	5.2.1	5.2.2	5.2.3
130.	Box closure not as specified.	5.2.1	5.2.2	5.2.3
131.	Gross weight exceeded as a result of quantity of like slings per box.	5.2.1	5.2.2	
132.	Box not modified with skids when required.	5.2.1	5.2.2	
133.	Marking not as specified.	5.3.1	5.3.1	5.3.2

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Each sling shall be preserved in accordance with MIL-P-116, method III.

5.1.2 Commercial. Each sling shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B or commercial, as specified (see 6.2).

5.2.1 Level A. Slings of like description, preserved as specified in 5.1, shall be packed in snug-fitting boxes conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. Box closure and strapping shall be in accordance with the applicable box specification or the appendix thereto except that the strapping shall be flat and the finish B. The gross weight shall not exceed 200 pounds except as follows:

> When the weight of a single sling is such that, when packed, it would exceed the 200 pound gross weight limitation, such slings, unless specified to be shipped loose (see 6.2), shall be individually packed in boxes as specified above provided each box is modified by the addition of skids in accordance with the applicable box specification.

5.2.2 Level B. The slings, preserved as specified in 5.1, shall be packed for level B the same as specified in 5.2.1 for level A except that the boxes shall be domestic type or class and the strapping shall be finish A or B.

5.2.3 Commercial. Slings of like description, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Level A and level B. In addition to any special or identification markings required by the contract or purchase order, including the specification part number, each bundle or container shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each bundle or container shall be marked with the weight and cube, and the specification part number.

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User Activities:
Army - AR
Navy - SH

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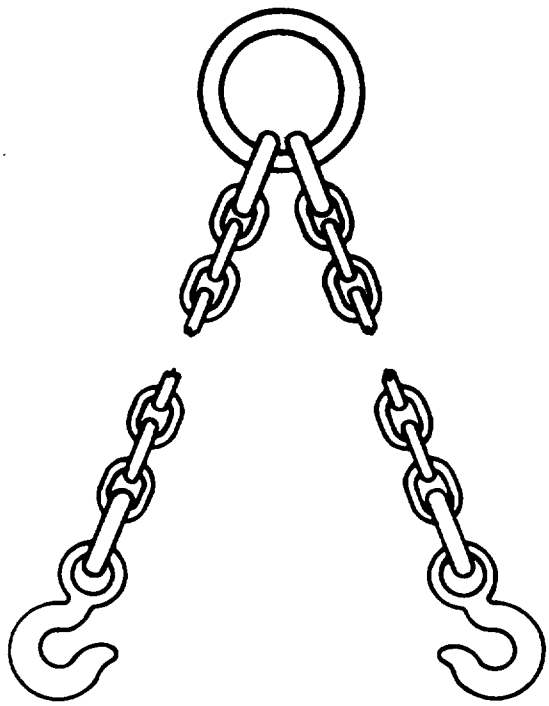


FIGURE 1. Class 1, 2 and 3 slings.

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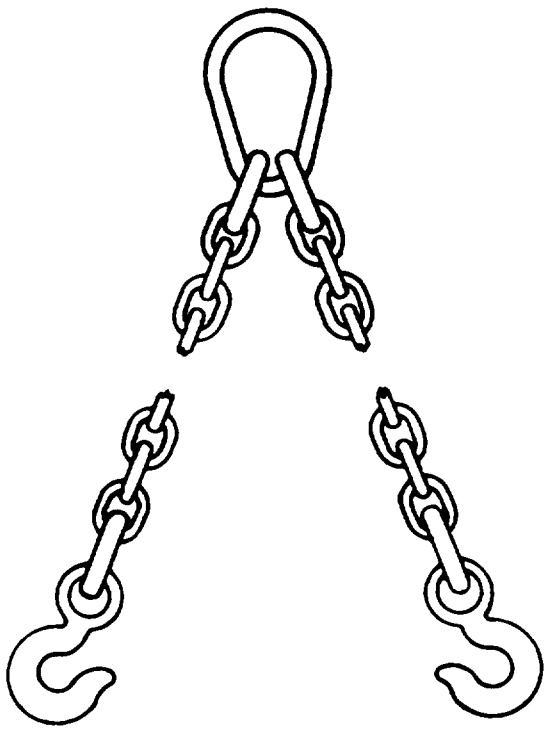


FIGURE 2. Class 4 and 5 slings.

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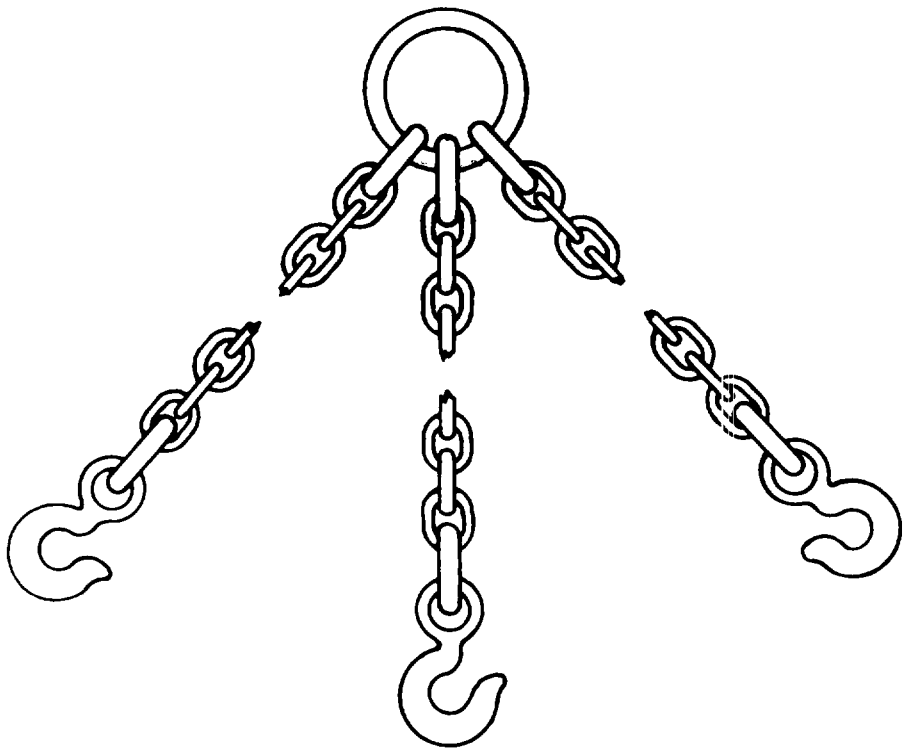
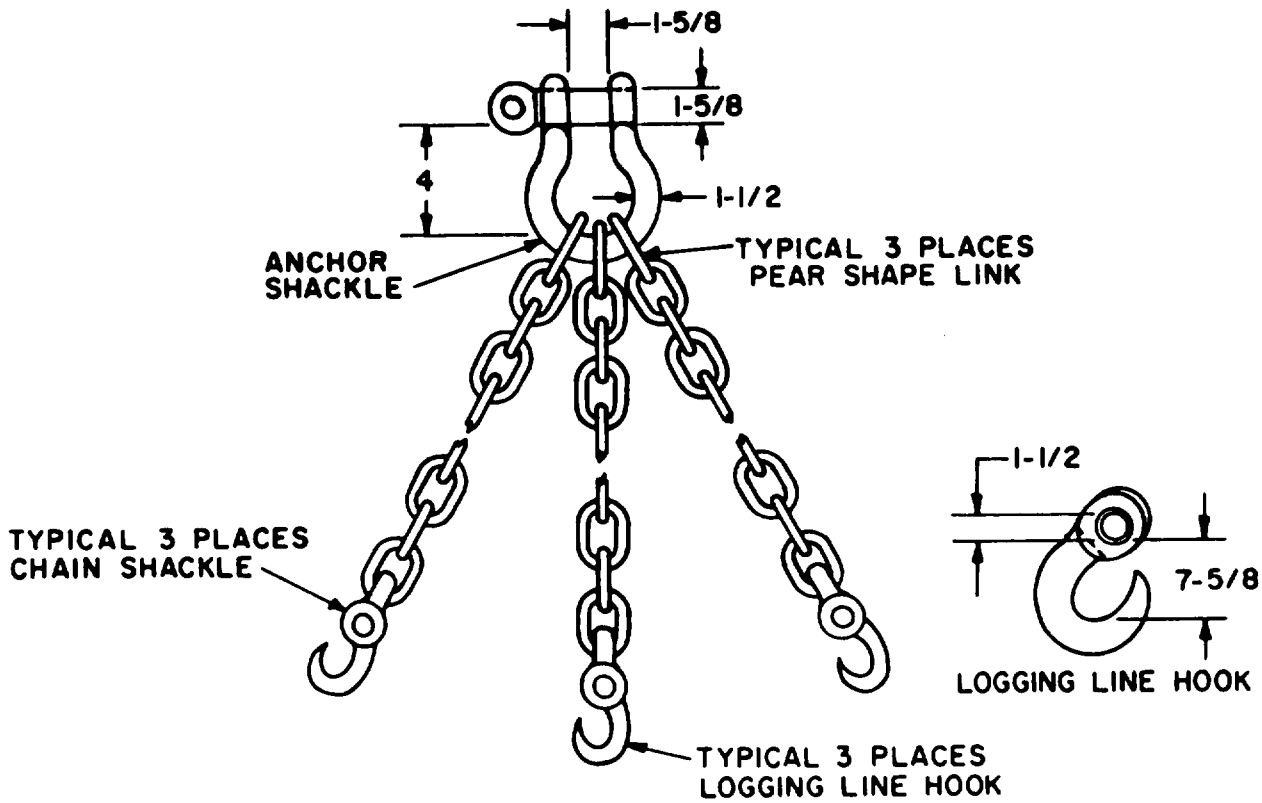


FIGURE 3. Class 6 sling.

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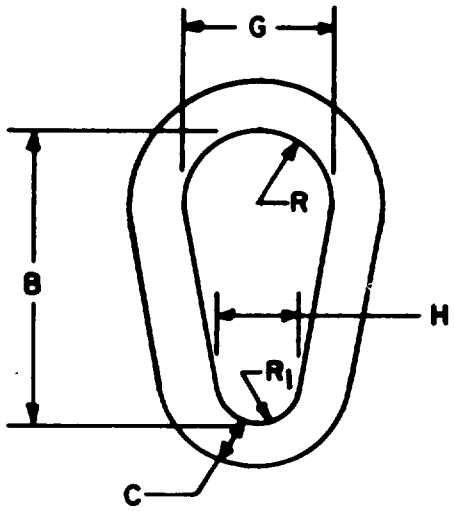


NOTE: ALL DIMENSIONS ARE IN INCHES.

FIGURE 4. Class 7 sling.

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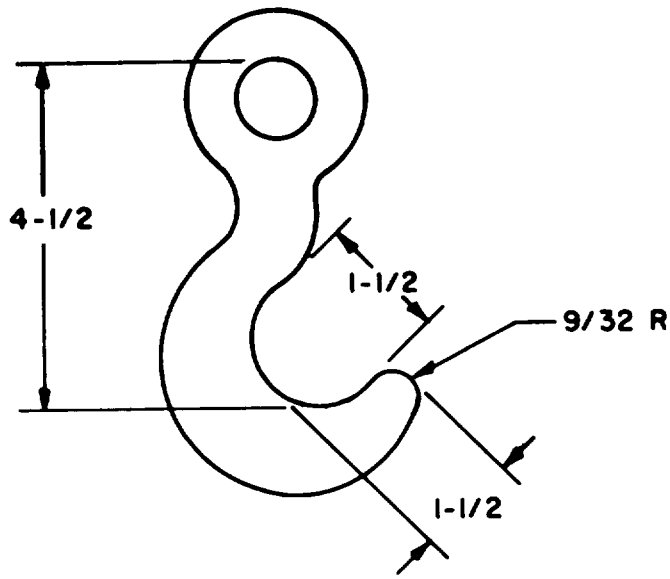
	CLASS 4	CLASS 5	CLASS 7
B	3	6	4-3/4
C	5/8	3/4	1
G	2	3	2-1/8
H	1-1/2	1-1/2	1-3/4
R	1	1-1/2	1-1/16
R_1	3/4	3/4	7/8

NOTE: ALL DIMENSIONS ARE IN INCHES.

FIGURE 5. Pear shaped link.

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NOTE: ALL DIMENSIONS ARE IN INCHES.

FIGURE 6. Drum hook.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-52432A	Slings, Multiple Leg (Chain)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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